

The Impact of a Health System Specialty Pharmacy Disease Management Program on Diabetes Mellitus

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**Southern Ohio
Medical Center**
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BACKGROUND

Diabetes type 1 and type 2 are diseases characterized by increased blood glucose caused by either insulin deficiency, insulin resistance, or both, which affect approximately one in ten Americans.

Type 2 diabetes affects approximately 34.2 million, or one of every ten, Americans. Diabetes has a high economic burden, accounting for an estimated \$327 billion including \$237 billion in direct medical expenses in the U.S. annually. For patients with an A1c greater than 7%, research has found that a reduction of A1c by 1% is associated with a significant decrease in all-cause diabetes related expenditures of \$736 annually.³

Adherence to medication regimens plays a key role in optimizing disease management and A1c reduction; however, barriers including cost, medication regimen complexity, and health literacy can prevent patients from adhering to their therapy regimens. Working with patients to overcome these barriers will reduce A1c and diabetes related expenditures.

Based on this information, Southern Ohio Medical Center (SOMC) in Portsmouth, OH launched a Chronic Disease Management (CDM) Pharmacy Service to help improve clinical outcomes for its patients with diabetes by providing high-touch support from onsite pharmacists and pharmacy liaisons. The pharmacists and liaisons work as integrated members of SOMC's Family Health Centers, collaborating with providers to monitor for adherence, identify and remove barriers to adherence, assess for therapy effectiveness, and recommend therapy changes as warranted.

All patients enrolled in the CDM Pharmacy Service receive medication access support, medication appropriateness and safety assessments, drug utilization review, adherence monitoring, as well as pharmacist counseling and education as needed to improve adherence. Patients considered high-risk based on their A1c value or a provider referral are enrolled in an education program to receive additional pharmacist counseling and support until they achieve their A1c goal.

OBJECTIVES

- Compare the reduction in hemoglobin A1c values for patients enrolled in CDM pharmacy services at SOMC compared to patients not enrolled in CDM pharmacy services at SOMC.
- Evaluate patient adherence to diabetes medications when managed by specialty pharmacy liaisons and pharmacists.

METHODS

Design Retrospective study of adult patients enrolled in CDM pharmacy services from July 2020 to March 2021 who had a baseline and at least one 3-month A1c completed. Patients were separated into three groups: high-risk patients with a baseline A1c > 10 who are enrolled in CDM pharmacy services, lower-risk patients with a baseline A1c <10 who are enrolled in CDM pharmacy services, and patients who choose not to enroll in CDM pharmacy services. Enrolled high-risk patients received A1c checks every 3 months, in addition to monthly glucose log reviews. Lower-risk patients received A1c checks every 6 months. Non-enrolled patients received only services rendered by their physician. All patients enrolled received proactive refill calls and compliance check-ins from the CDM pharmacy team.

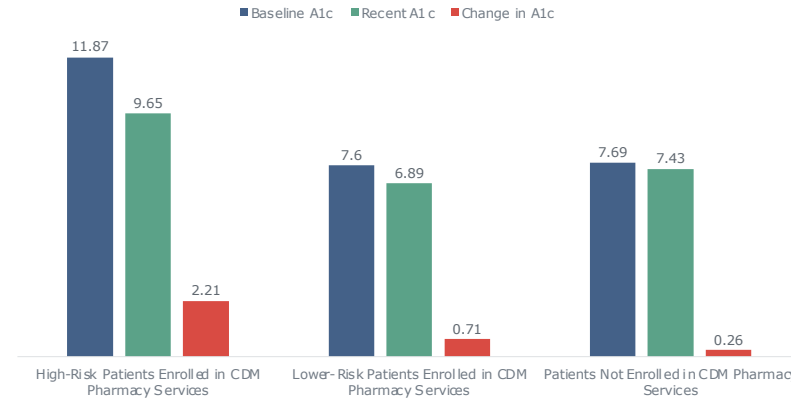
Setting Outpatient family practice offices at Southern Ohio Medical Center (SOMC), a community medical center serving rural Ohio and Kentucky.

Measures Patient A1c lab values, date of enrollment into CDM pharmacy services, date of enrollment into additional support program for high-risk patients, date of graduation from additional support program for high-risk patients

Analysis Average reduction of A1c from baseline to most recent follow A1c that is at least three months time has elapsed since baseline A1c was collected for all enrolled patients. Average time from enrollment to graduation from education program for high-risk patients.

RESULTS

Hemoglobin A1c of Different Patient Groups



Graduation Data for High-Risk Patients

	4Q2020	1Q2021
Patients Graduated	2	11
Average Baseline A1c (%)	9.5	10.4
Average Current A1c (%)	7	6.9
Time to Graduation (days)	70.5	110
Delta A1c (%)	2.5	3.52

Estimated Annual Cost Savings (\$)

Patients Not Enrolled in CDM Pharmacy Services	191
Lower-Risk Patients Enrolled in CDM Pharmacy Services	522
High-Risk Patients Enrolled in CDM Pharmacy Services and Education Program	1619

CONCLUSIONS

- Effects of Services on A1c**
 - Patients enrolled had greater reduction in A1c compared to those non-enrolled, and high-risk patients receiving additional education and support had the greatest decreases
 - High-risk patients enrolled in education program had an average decrease of 2.21
 - Lower-risk patients enrolled in CDM pharmacy services had a decrease from baseline of 0.71
 - Non-enrolled patients had a decrease of 0.26
- Cost Savings Related to A1c Reduction**
 - 1% A1c reduction equates to savings of \$736 per year³
 - 2.2% A1c reduction would equate to \$1619 per year or \$135 per month in cost savings from diabetes-related expenditures in SOMC's patient population
- Medication Adherence**
 - PDC for patients enrolled in CDM pharmacy services is 0.91
 - Average national PDC for diabetes medications 0.36-0.81⁵
- Next Steps**
 - Expansion into additional family health clinics
 - Continued outreach to patients with diabetes for enrollment in CDM pharmacy services
 - Continued expansion into other disease states
 - Asthma, COPD, heart failure, hyperlipidemia, atrial fibrillation

REFERENCES

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